Class 9 Mathematics — Exercise 3.2

Basic Concept

Number of Elements of a Set

$n(A)$ means the **number of elements** in set A.

Important Formulas

For two sets:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(B) = n(A \cap B) + n(B - A), \qquad n(A \cup B) = n(A - B) + n(A \cap B) + n(B - A)$$

For disjoint sets (no common element): $n(A \cup B) = n(A) + n(B)$

For three sets:

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

Neither / None: $n(\text{none}) = n(U) - n(A \cup B)$

Method for Venn diagram: Fill the **innermost (common) region first**, then subtract to get the "only" regions.

Question 1

Let A and B be two finite sets such that $n(A) = 24$, $n(B) = 18$ and $n(A \cup B) = 31$. Find $n(A \cap B)$.

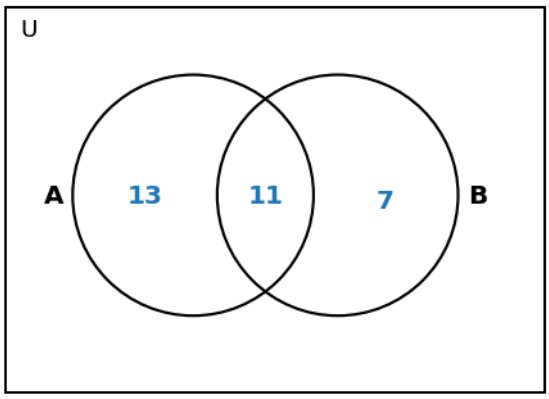
$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$31 = 24 + 18 - n(A \cap B)$$

$$31 = 42 - n(A \cap B)$$

$$n(A \cap B) = 42 - 31$$

$$\boxed{n(A \cap B) = 11}$$

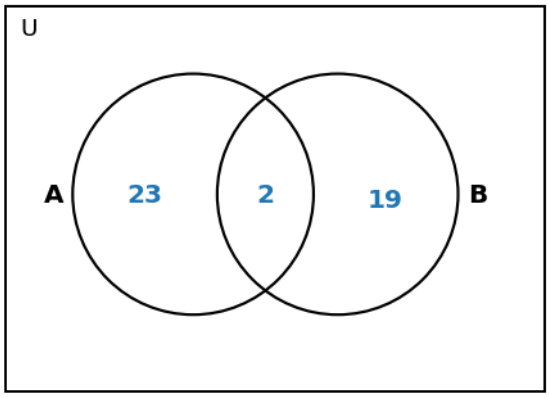


Question 2

If $n(A-B)=23$, $n(A\cup B)=44$ and $n(A\cap B)=2$, then find $n(B-A)$. Also find $n(B)$. (Hint: $n(B)=n(A\cap B)+n(B-A)$)

$$n(A\cup B)=n(A-B)+n(A\cap B)+n(B-A)$$
$$44=23+2+n(B-A)$$
$$n(B-A)=44-25$$
$$\boxed{n(B-A)=19}$$

$$n(B)=n(A\cap B)+n(B-A)=2+19$$
$$\boxed{n(B)=21}$$



Question 3

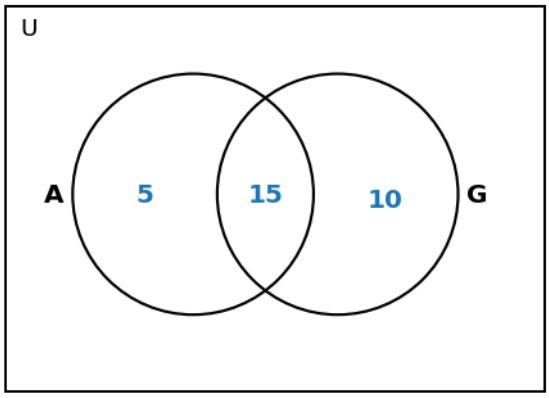
In a group of 30 Mathematics students, 20 like Algebra and 15 like both Geometry and Algebra. Show the data by Venn diagram. Also find how many students like Geometry.

Total students $=n(A\cup G)=30,\quad n(A)=20,\quad n(A\cap G)=15,\quad n(G)=?$

$$n(A\cup G)=n(A)+n(G)-n(A\cap G)$$
$$30=20+n(G)-15$$
$$30=5+n(G)$$

$$\boxed{n(G) = 25}$$

Only Algebra $= 20 - 15 = 5$, Only Geometry $= 25 - 15 = 10$



Question 4

In a street with 50 houses, 25 houses have lawns, 32 houses have car porch and 15 houses have both lawn and car porch. Show the data by Venn diagram. Also find how many houses have neither lawn nor porch.

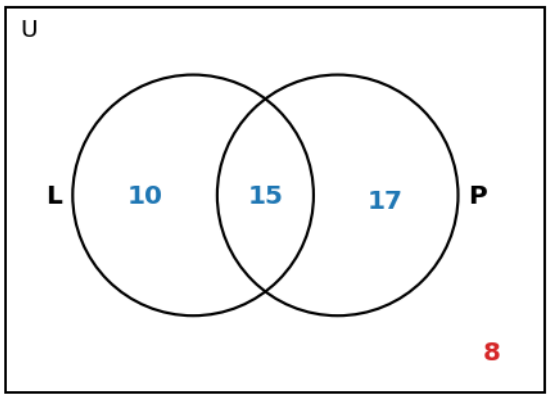
$n(U) = 50, \quad n(L) = 25, \quad n(P) = 32, \quad n(L \cap P) = 15$

$$n(L \cup P) = n(L) + n(P) - n(L \cap P) = 25 + 32 - 15 = 42$$

$$\text{Neither} = n(U) - n(L \cup P) = 50 - 42$$

$$\boxed{8 \text{ houses}}$$

Only lawn $= 25 - 15 = 10$, Only porch $= 32 - 15 = 17$



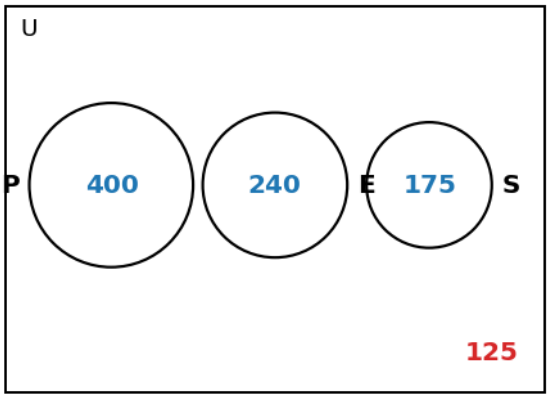
Question 5

In a survey of 940 children, 400 students were found studying at primary level, 240 at elementary and 175 at secondary level. Create a Venn diagram to illustrate this information. How many children were found out of school?

A child studies at only one level, so the sets P, E and S are **disjoint**.

$$n(P \cup E \cup S) = n(P) + n(E) + n(S) = 400 + 240 + 175 = 815$$

Out of school $= 940 - 815 =$ $\boxed{125 \text{ children}}$



Question 6

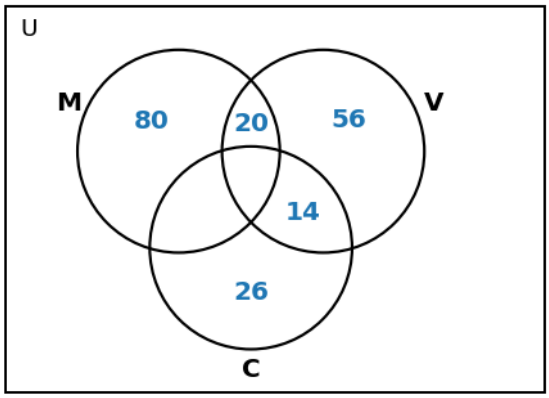
ABC Dairy polls its customers on their favorite flavor: chocolate, vanilla or mango. 100 customers said they like mango flavor, 90 customers said they like vanilla, 40 polled for chocolate, 20 customers liked both mango and vanilla while 14 liked both chocolate and vanilla. How many customers said they like: (i) only mango (ii) only vanilla (iii) only chocolate?

$n(M) = 100,\ \ n(V) = 90,\ \ n(C) = 40,\ \ n(M \cap V) = 20,\ \ n(C \cap V) = 14$

(i) Only mango $= 100 - 20 =$ $\boxed{80}$

(ii) Only vanilla $= 90 - 20 - 14 =$ $\boxed{56}$

(iii) Only chocolate $= 40 - 14 =$ $\boxed{26}$



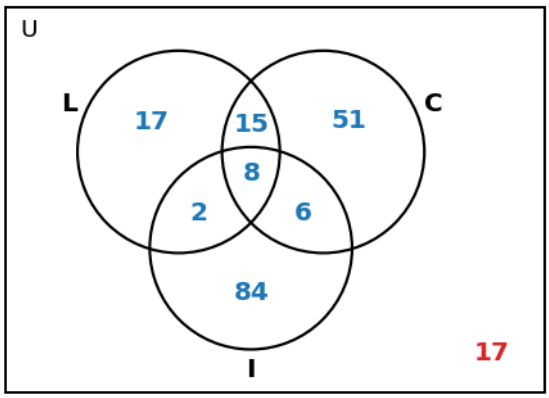
Question 7

In a survey of university 200 students were interviewed. It was found that: 42 students have laptops, 80 students have cell phones, 100 students have iPods, 23 students have both a laptop and a cell phone, 10 students have both a laptop and iPod, 14 students have both a cell phone and iPod and 8 students have all three items. (a) How many students have only cell phone? (b) How many students have none of the three items? (c) How many students have both iPod and laptop but not cellphone?

Total $= 200;\ \ n(L) = 42,\ \ n(C) = 80,\ \ n(I) = 100;\ \ n(L \cap C) = 23,\ \ n(L \cap I) = 10,$ $n(C \cap I) = 14,\ \ n(L \cap C \cap I) = 8$

Filling the Venn diagram from the centre:

- $L \cap C$ only $= 23 - 8 = 15$, $L \cap I$ only $= 10 - 8 = 2$, $C \cap I$ only $= 14 - 8 = 6$
- Only laptop $= 42 - (15 + 8 + 2) = 17$
- Only cell phone $= 80 - (15 + 8 + 6) = 51$
- Only iPod $= 100 - (2 + 8 + 6) = 84$



(a) Only cell phone $= 80 - (15 + 8 + 6) = 80 - 29 = \boxed{51}$

(b) $n(L \cup C \cup I) = 42 + 80 + 100 - 23 - 10 - 14 + 8 = 183$

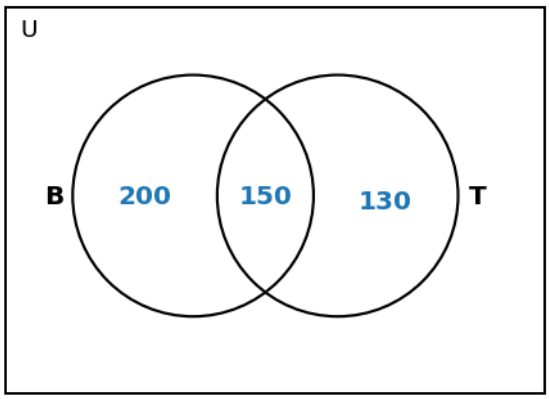
$$\text{None} = 200 - 183 = \boxed{17}$$

(c) Both iPod and laptop but not cell phone $= n(L \cap I) - n(L \cap C \cap I) = 10 - 8 = \boxed{2}$

Question 8

In a girls college, every student plays either badminton or table tennis or both. If 350 students play badminton, 280 play table tennis and 150 play both, find how many students are there in the college.

$$n(B \cup T) = n(B) + n(T) - n(B \cap T) = 350 + 280 - 150 = \boxed{480 \text{ students}}$$

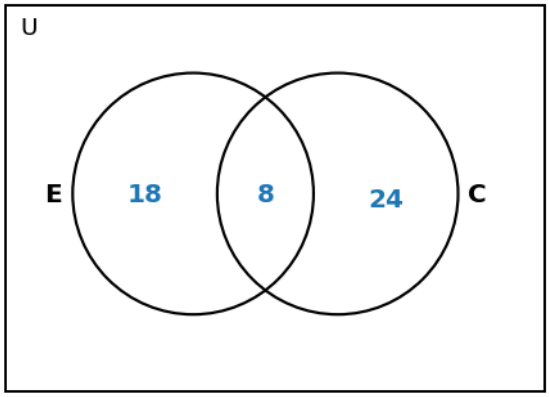


Question 9

Among 50 students, 8 are learning both English and Chinese languages. A total of 26 students are learning English. If every student is learning at least one language, how many students are learning Chinese?

$n(E \cup C) = 50$, $n(E) = 26$, $n(E \cap C) = 8$

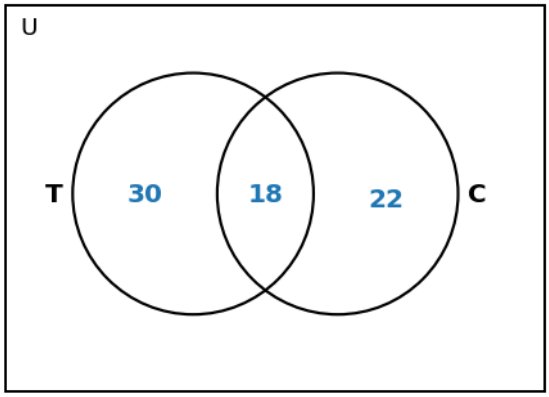
$$n(E \cup C) = n(E) + n(C) - n(E \cap C)$$
$$50 = 26 + n(C) - 8$$
$$\boxed{n(C) = 32 \text{ students}}$$



Question 10

Out of 70 people, 48 like tea and 40 like coffee and each person likes at least one of the two drinks. How many like both tea and coffee?

$$n(T \cup C) = n(T) + n(C) - n(T \cap C)$$
$$70 = 48 + 40 - n(T \cap C)$$
$$\boxed{n(T \cap C) = 18 \text{ people}}$$

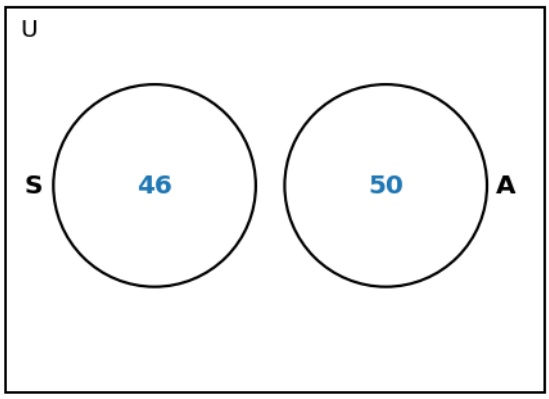


Question 11

There are 46 students in science group and 50 students in arts group. Find the number of students who are either in science or arts group.

A student is in only one group, so the sets are **disjoint**:

$$n(S \cup A) = n(S) + n(A) = 46 + 50 = \boxed{96 \text{ students}}$$



Question 12

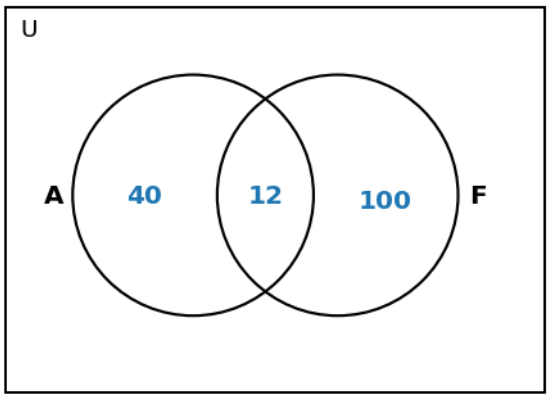
In a group of people, 52 people can speak Arabic and 112 can speak French. How many can speak Arabic only? How many can speak French only if 12 of them can speak both languages? How many people were in the group?

$n(A) = 52, \quad n(F) = 112, \quad n(A \cap F) = 12$

Arabic only $= 52 - 12 = \boxed{40}$

French only $= 112 - 12 = \boxed{100}$

$$n(A \cup F) = n(A) + n(F) - n(A \cap F) = 52 + 112 - 12 = \boxed{152 \text{ people}}$$



Question 13

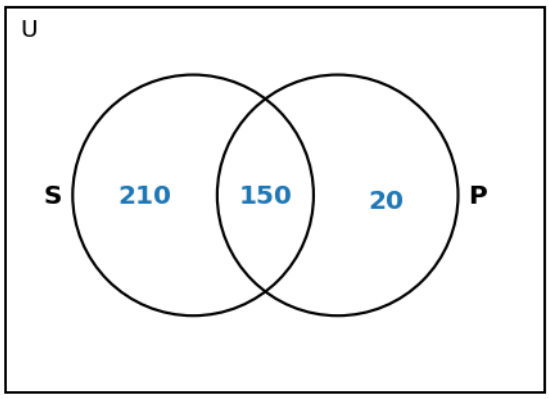
In a high school, 360 students like reading story books, 170 like practical activities and 150 like both. Find (i) the number of students who like reading story books only (ii) the number of students who like only practical activities (iii) the total number of students in the school.

$n(S) = 360, \quad n(P) = 170, \quad n(S \cap P) = 150$

(i) Story books only $= 360 - 150 = \boxed{210}$

(ii) Practical activities only $= 170 - 150 = \boxed{20}$

(iii) $n(S \cup P) = 360 + 170 - 150 = \boxed{380}$



Question 14

In a survey of 60 people, it was found that 25 people watch channel A, 16 watch channel B, 13 watch channel C, 4 watch both A and B, 7 watch both B and C, 8 watch both A and C, 3 watch all three channels. Find the number of people who watch at least one of the channels. Also find number of people who do not watch these channels.

$n(T) = 60;\ \ n(A) = 25,\ \ n(B) = 16,\ \ n(C) = 13;\ \ n(A \cap B) = 4,\ \ n(B \cap C) = 7,\ \ n(A \cap C) = 8,$
$n(A \cap B \cap C) = 3$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$= 25 + 16 + 13 - 4 - 7 - 8 + 3$$

$$= \boxed{38 \text{ people watch at least one channel}}$$

$$\text{Do not watch} = 60 - 38 = \boxed{22 \text{ people}}$$

